Line Through Points

AlgoExpert · Very Hard · Arrays

The problem

- Maximum number of the given (unique, integer) points lying on one straight line.
- For each anchor, count normalised slopes (reduced by gcd, sign-canonicalised).

Intuition

- Fix an anchor point; points on the same line through it share a normalised slope.

Brute force vs optimal

- Every pair defines a line, count points on it → $O(n^3)$
- Anchor + slope hash map → $O(n^2)$
- Reduce slopes by gcd to avoid floats

Approach

- 1. For each anchor, map slopes to counts
- 2. Normalise slopes with gcd and sign
- 3. Track the largest count plus one

Worked example

slope dy/dx	points
1/1	2
-3/1	1
-5/3	1
-1/3	1
0/1	1

```
anchor (1, 1): best line has 3 points
overall best 3
```

Anchor (1, 1)

Worked example

slope dy/dx	points
1/1	1
-1/1	3
1/0	1

```
anchor (2, 2): best line has 4 points
overall best 4
```

Anchor (2, 2)

Worked example

slope dy/dx	points
-1/3	1
-3/5	1
-3/1	1
2/1	1

```
anchor (3, 3): best line has 2 points
overall best 4
```

Anchor (3, 3)

Java solution

```java
public class LineThroughPoints {
    public static int lineThroughPoints(int[][] points) {
        int best = Math.min(points.length, 1);
        for (int i = 0; i < points.length; i++) {
            Map<String, Integer> slopes = new HashMap<>();
            for (int j = i + 1; j < points.length; j++) {
                int dx = points[j][0] - points[i][0], dy = points[j][1] - points[i][1];
                int g = gcd(Math.abs(dx), Math.abs(dy));
                dx /= g; dy /= g;
                if (dx < 0 || (dx == 0 && dy < 0)) { dx = -dx; dy = -dy; }
                best = Math.max(best, slopes.merge(dx + "/" + dy, 1, Integer::sum) + 1);
            }
        }
        return best;
    }

    private static int gcd(int a, int b) { return b == 0 ? a : gcd(b, a % b); }
}
```

Complexity

- Time: O(n^2)
- Space: O(n)

Edge cases & pitfalls

- Use reduced (dx, dy), not floats
- Normalise sign
- Vertical lines

Interview tips & follow-ups

- Mention duplicates handling

Recap

- Fix an anchor point; points on the same line through it share a normalised slope.
- Complexity: O(n^2)